

Topology-Synchronous Decoupled Observations and Supervised-Reset Estimation for Online C-ESR Monitoring of the Ćuk Energy-Transfer Capacitor

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Abstract—The energy-transfer capacitor in a Ćuk converter is a reliability-critical power-transfer element, but ripple measurements couple its capacitance and equivalent series resistance (ESR). This paper proposes topology-synchronous, direction-decoupled observations that exploit the current commutation between $+i_{L1}$ and $-i_{L2}$ without diagnostic injection: timestamped fits in guarded edge windows reconstruct an ESR-dominant discontinuity, and in-state charge integration yields a capacitance-dominant row. Positive finite-window information bounds establish local identifiability under valid continuous-conduction-mode conditions. On these observations, a supervised-reset Kalman estimator (TS-SRKE) augments the Joseph-form kernel with a predeclared two-time-scale supervisor over accepted and gate-rejected innovations, restoring gain after abrupt health changes without disturbing interval calibration. Evaluation covers 48 blind cases across four noise profiles and four timing offsets, with abrupt-step, degradation-ramp, and TMS320F28379D device-realistic scenarios. Simulation results show that the proposed observations alone reduce mean capacitance error from 7.0366% to 0.3727% with the RLS kernel unchanged, and that TS-SRKE attains 0.3011%/0.6446% capacitance/ESR error with 88.93% joint coverage while reproducing its parent exactly in steady state. Abrupt capacitance and ESR steps are settled within 1.52 and 0.48 ms, where the unsupervised kernel does not recover; worst 95th-percentile errors of 1.6425%/2.6109% confirm feasibility within a constrained DSP budget.

Index Terms—Ćuk converter, capacitor condition monitoring, change detection, energy-transfer capacitor, capacitance estimation, equivalent series resistance, Kalman filters, parameter identification, power electronics reliability.

I. INTRODUCTION

Capacitance loss and equivalent series resistance (ESR) growth alter electrical stress, thermal loss, and filtering performance in power converters, and both are established degradation precursors for electrolytic and film capacitors [1], [2]. Online monitoring methods therefore span ripple analysis, injected excitation, circuit reconfiguration, observers, recursive identification, and data-driven processing [3]–[5]. Their applicability depends on the capacitor’s circuit role and on which electrical variables are already measured.

The energy-transfer capacitor in a Ćuk converter is distinct from a dc-link or output filter: it lies in the main power-transfer path and alternately carries $+i_{L1}$ and $-i_{L2}$. Normal switching thus creates a bidirectional current commutation that can expose capacitor health without an added diagnostic perturbation. Treating this component as a generic two-terminal capacitor sampled asynchronously discards that topology-specific information.

Prior joint C-ESR methods use inherent ripple or within-cycle relations [6]–[8], switched networks and observers [9]–[11], or inherent and injected spectral features [12]–[14]. Related recent work includes sensorless boost ESR estimation, optimized buck-capacitor injection, and large-signal C-ESR estimation [15]–[17]. Experimentally validated dc-link monitoring continues to advance through maximum-likelihood traction-inverter estimation, adaptive-forgetting RLS for cascaded-H-bridge converters, discharge-profile and grey-box transient methods, multilevel-converter ESR-C schemes, and DSP-efficient spectral precursors [18]–[24]. Wavelet/Kalman and recursive RLS/Kalman formulations provide further signal-processing and estimation routes [25]–[28]. Ćuk-specific modeling, fault diagnosis, and prognostics have also been studied [29]–[32]. These studies establish the broader tools; the gap addressed here is how the Ćuk transfer mechanism itself can be separated into implementable edge and charge information.

This paper reconstructs the switching-edge discontinuity for an ESR-dominant observation and uses an ESR-compensated in-state charge balance for a capacitance-dominant observation. Table I positions that scope against representative methods. The comparison is categorical rather than numerical because the converter topologies, sensing sets, and reporting protocols differ across studies.

The contributions are fivefold.

- 1) The normal Ćuk switching sequence is interpreted as topology-induced health excitation through the $+i_{L1} \leftrightarrow -i_{L2}$ capacitor-current commutation.
- 2) Timestamp-reconstructed edge and safe-window charge observations provide ESR-dominant and capacitance-dominant information while preserving the distinction between terminal and internal capacitor voltage.
- 3) Positive finite-window information bounds establish local identifiability under valid continuous-conduction-mode conditions.
- 4) A supervised-reset estimator (TS-SRKE) augments the structured Kalman kernel with a predeclared two-time-scale mean-shift supervisor over the accepted *and* gate-rejected normalized innovations, restoring gain after abrupt health changes without disturbing steady-state behavior or interval calibration. The components—EWMA residual filtering, change supervision, and covariance resetting—are classical [33]–[36]; the contribution is their direction-specific construction on gated topology-synchronous rows, and neither the RLS nor the Kalman

TABLE I
REPRESENTATIVE RELATED METHODS AND THE SCOPE OF THIS WORK

Method	Capacitor role	Joint C/ESR	Added excitation	Measurements	Processing	Distinction from this work
Hasegawa <i>et al.</i> [6]	inverter dc link	yes	no	dc-link voltage and ripple current	ripple-frequency separation	different topology and ripple bands
Yao <i>et al.</i> [7]	buck output filter	yes	no	PWM timing and output voltage	within-cycle relation	topology-specific relation without Ćuk current reversal
Barbulescu <i>et al.</i> [9]	device under test	yes	switched network	terminal voltage and switched resistive network	parameter observer	requires an additional discharge network
Ribeiro <i>et al.</i> [12]	NPC dc link	yes	no	converter voltage/current signals	spectral features plus RLS	different inherent-signal mapping
Amaral <i>et al.</i> [25]	boost output/dc link	yes	no	converter signals	wavelet plus Kalman filtering	reconstructed frequency features rather than edge/charge rows
This work	Ćuk energy transfer	yes	no	v_T , i_1 , i_2 , PWM state/time	edge fit and charge balance	uses $+i_{L1} \leftrightarrow -i_{L2}$ commutation

Sensing note: No diagnostic injection does not imply zero sensing burden. The device-realistic realization uses separate absolute- and edge-voltage ranges in the differential v_T analog front end, but no capacitor-branch current sensor.

recursion is claimed as a novelty.

- 5) A common evaluation protocol combines independent plant traces, 48 blind cases, noise/timing sweeps, an observation–estimator factorial, degradation trajectories, operating transients, closed-loop-regulation and light-load-availability studies, a Cramér–Rao lower bound (CRLB) efficiency analysis, and a TMS320F28379D device-realistic simulation.

The remainder of the paper develops the model, observations, and finite-window bounds, then presents the supervised estimator and evaluates it against its unsupervised baselines.

II. ENERGY-TRANSFER CAPACITOR MODEL AND HEALTH SIGNATURES

A. Topology, States, and Sign Convention

Consider the classical inverting Ćuk converter in continuous conduction mode (CCM). The state vector is

$$\mathbf{x} = [i_1 \quad i_2 \quad v_C \quad v_o]^\top, \quad (1)$$

where $i_1 = i_{L1} > 0$ and $i_2 = i_{L2} > 0$ follow their usual inductor-current directions, v_C is the internal voltage of the ideal element in the energy-transfer capacitor, and $v_o > 0$ is the magnitude of the physically negative output voltage. Let $u \in \{0, 1\}$ denote the main-switch state, with $u = 1$ for the on-state.

The transfer-capacitor current is

$$i_C = (1 - u)i_1 - ui_2, \quad (2)$$

and therefore

$$u = 0 : i_C = i_1, \quad u = 1 : i_C = -i_2. \quad (3)$$

Equation (2) is the topology-specific reconstruction used throughout this work. It avoids a dedicated series current sensor in the capacitor branch, provided that the switching state and both inductor currents are available.

TABLE II
PRINCIPAL NOTATION

Symbol	Meaning
u, D, T_s	switch state, duty ratio, switching period
i_1, i_2	input- and output-inductor currents
i_C	reconstructed energy-transfer-capacitor current
v_C, v_T	ideal internal and measured terminal voltages
C, C_b	actual and fixed nominal capacitances
r_C	capacitor ESR at the measurement conditions
$\bar{\alpha}$	normalized inverse capacitance, C_b/C
q	charge transferred inside an accepted window
I_Σ	edge-current sum, $i_1 + i_2$
k_R	training-calibrated gain held fixed during estimation

B. Nonideal Capacitor Port

The first-order capacitor model contains an ideal capacitance C in series with ESR r_C :

$$v_T = v_C + r_C i_C, \quad \dot{v}_C = \frac{i_C}{C}, \quad (4)$$

where v_T is the externally measured terminal voltage. Confusing v_T with v_C would attribute the ESR drop to the ideal capacitance and couple the two health indicators.

For numerical conditioning, a fixed nominal capacitance C_b defines

$$\bar{\alpha} = \frac{C_b}{C}, \quad C = \frac{C_b}{\bar{\alpha}}. \quad (5)$$

The health-parameter vector is $\boldsymbol{\theta} = [\bar{\alpha}, r_C]^\top$.

C. Health Scope

Capacitance loss and ESR growth are treated as condition indicators, not as remaining-useful-life predictions. Because r_C depends on temperature, frequency, operating point, and aging, the method estimates effective ESR under the specified switching and acquisition conditions. The theory assumes recurring valid CCM windows, negligible parameter change within an identification window, and a first-order series C–ESR model. Equivalent series inductance (ESL), ringing, discontinuous conduction mode (DCM), and front-end nonidealities are

TABLE III
CONVERTER AND IDENTIFICATION PARAMETERS

Item	Value
Input voltage	$V_{in} = 24$ V nominal; 19.2/24/28.8 V blind range
Duty ratio	$D = 0.40$ nominal; 0.30/0.40/0.55/0.65 blind range
Switching frequency	$f_s = 50$ kHz ($T_s = 20$ μ s)
Inductors	$L_1 = L_2 = 500$ μ H
Capacitors	$C_b = 100$ μ F; $C_o = 470$ μ F
Nominal ESR/load	$r_b = 50$ m Ω ; $R = 10$ Ω
Primary recursion	$\lambda = 0.9975$; $P_0 = 1000$
Edge calibration	$k_R = 0.9772$; training-calibrated nominal value; case-specific cali- brated value used in evaluation
C charge window	$T_{w,C} = 2.0$ μ s
ESR edge-fit win- dow	$T_{w,R} = 2.2$ μ s in the F28379D realization
Health reporting	1024 cycles = 20.48 ms per external report

addressed through safe-window selection, calibration, gating, or explicit limitations.

Only inverse capacitance is normalized: $\bar{\alpha} = C_b/C$ is dimensionless, whereas r_C is estimated directly in ohms. Separate scalar recursions and physical bounds avoid mixing their numerical scales. The predeclared ESR interval is $0.35r_b \leq r_C \leq 2.50r_b$, or 17.5–125 m Ω for $r_b = 50$ m Ω .

III. TOPOLOGY-SYNCHRONOUS DECOUPLED OBSERVATION FRAMEWORK

A. Natural Bidirectional Excitation

At a valid $u : 0 \rightarrow 1$ transition, the capacitor current changes from $i_C^- = i_1^-$ to $i_C^+ = -i_2^+$. Hence

$$\Delta i_C = i_C^+ - i_C^- = -(i_1^- + i_2^+). \quad (6)$$

Because the internal ideal-capacitor voltage cannot jump, $v_C^+ = v_C^-$. The terminal voltages immediately before and after the edge satisfy

$$v_T^- = v_C + r_C i_1^-, \quad (7)$$

$$v_T^+ = v_C - r_C i_2^+, \quad (8)$$

which gives the ideal ESR signature

$$v_T^- - v_T^+ = r_C (i_1^- + i_2^+). \quad (9)$$

Thus, the topology transition provides a large, signed current change while cancelling the continuous internal voltage.

B. Timestamp-Reconstructed ESR Observation

Direct edge samples are contaminated by acquisition skew, finite bandwidth, and switching transients. The proposed measurement therefore fits local linear models to timestamped points in safe pre- and post-edge windows and extrapolates both models to a common switching timestamp. With reconstructed intercepts $\hat{v}_T^-$ and $\hat{v}_T^+$, the accepted ESR row is

$$z_R = \hat{v}_T^- - \hat{v}_T^+ = k_R I_{\Sigma} r_C + \nu_R, \quad I_{\Sigma} = i_1^- + i_2^+, \quad (10)$$

where k_R is a training-locked calibration coefficient and ν_R collects voltage, timestamp, fitting, and residual front-end errors. The reconstruction estimates the discontinuity at one physical instant instead of subtracting samples taken at different offsets.

Simulation calibration and blind health identification use disjoint cases. Four calibration profiles—nominal, high duty/load, high noise, and nominal parasitic Model B—form the accepted-edge ratio

$$\kappa_m = \frac{z_{R,m}}{I_{\Sigma,m} r_{C,\text{ref},m}}, \quad \hat{k}_R = \text{median}_m \kappa_m, \quad (11)$$

where $r_{C,\text{ref},m}$ is the planted simulation reference ESR. The resulting training-locked coefficient is $k_R = 0.9772$. The manuscript reports k_R to four decimal places; the full-precision value used in all regenerated results is retained in Supplementary Note S2 and the source-traceability record.

In the evaluation, a declared ± 0.006 case-wise spread about $\hat{k}_R$ represents unit-to-unit variation in the calibrated measurement-chain gain. For each simulated case, the realized coefficient is assumed known after calibration and is supplied identically to all compared estimators through $h_R = k_R I_{\Sigma}$. It therefore tests calibrated unit-to-unit variability rather than residual unknown calibration error; k_R is never updated from blind residuals. A hardware realization would require commissioning with an independently characterized reference capacitor under the final measurement chain.

Thus, k_R is known and fixed within each identification interval; otherwise the edge row identifies only the product $k_R r_C$. The smaller evaluated-envelope value $k_{R,\text{min}}^{\text{eval}} = 0.957654$ is used only for a conservative information bound and does not resolve this scale ambiguity.

C. Safe-Window Charge Observation

For an interval $[t_a, t_b]$ wholly inside one switching state, with the charge-safe-window length denoted by $T_{w,C} = t_b - t_a$, define

$$q = \int_{t_a}^{t_b} i_C(t) dt. \quad (12)$$

Integration of (4) yields

$$\Delta v_T = \frac{q}{C} + r_C \Delta i_C. \quad (13)$$

Using the most recent accepted ESR estimate gives the capacitance pseudo-measurement

$$z_C = \Delta v_T - \hat{r}_C \Delta i_C = \frac{q}{C_b} \bar{\alpha} + \nu_C. \quad (14)$$

The charge row is evaluated away from the switching edge and is capacitance-dominant after ESR compensation. ESR-estimation uncertainty and residual correlation enter the conditional measurement covariance.

More explicitly, first-order substitution of measured end-point current, voltage, charge, and the preceding ESR estimate gives

$$z_C = \frac{q}{C_b} \bar{\alpha} + (r_C - \hat{r}_C) \Delta i_C + \nu_{\Delta v} - \hat{r}_C \nu_{\Delta i} + \nu_q, \quad (15)$$

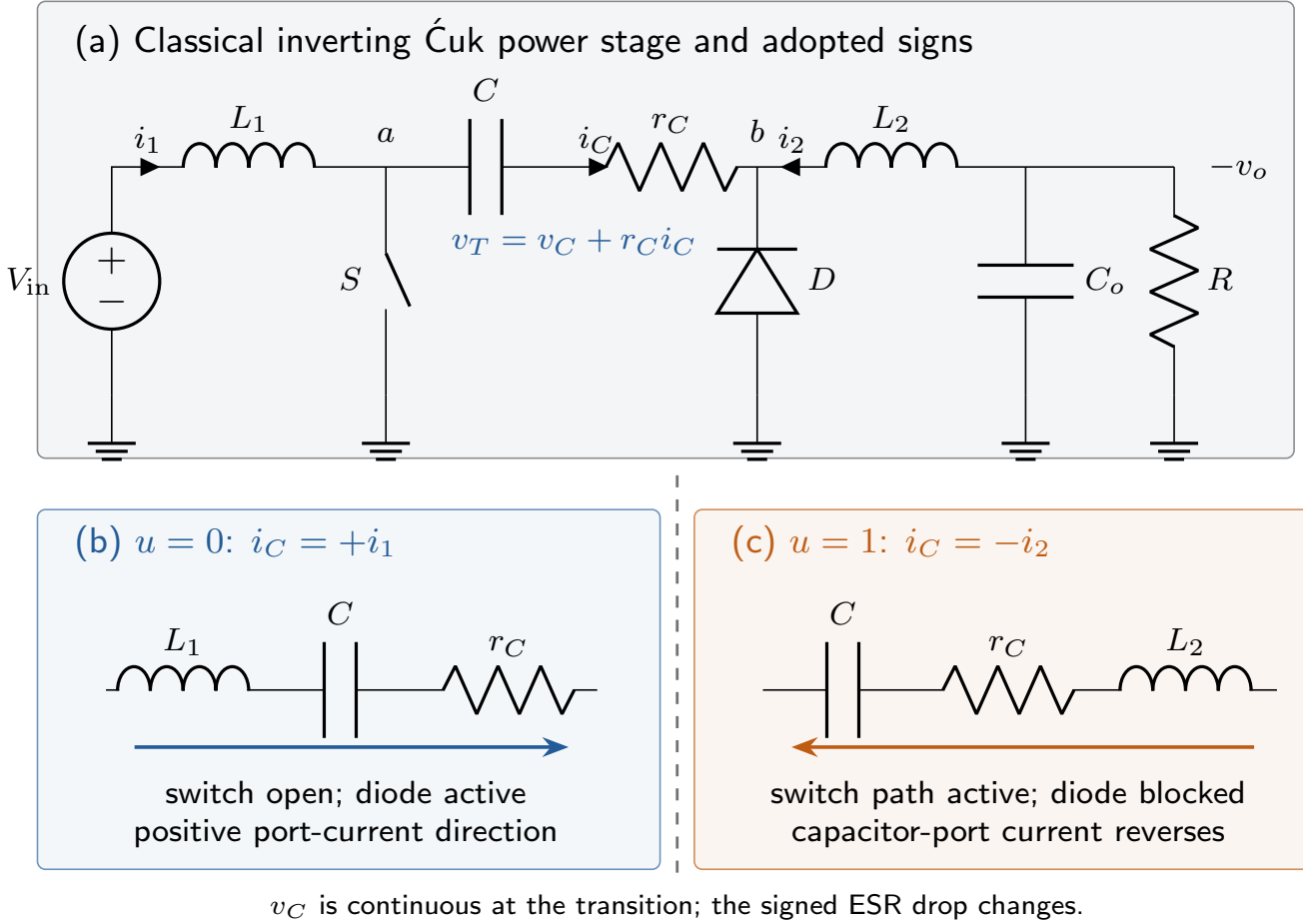


Fig. 1. Classical inverting Ćuk converter and adopted current/voltage signs: (a) full power stage; (b) $u = 0$ topology with $i_C = +i_1$; (c) $u = 1$ topology with $i_C = -i_2$. The ideal capacitor voltage v_C is continuous at a transition, while the terminal voltage $v_T = v_C + r_C i_C$ contains the ESR step.

TABLE IV
DECOUPLED OBSERVATION ROWS

Row	Measurement	Regression	Dominant parameter
Edge	$\hat{v}_T^- - \hat{v}_T^+$	$k_R I_{\Sigma} r_C + \nu_R$	r_C
Charge	$\Delta v_T - \hat{r}_C \Delta i_C$	$(q/C_b) \bar{\alpha} + \nu_C$	$\bar{\alpha}$

where ν_q is the voltage-equivalent charge-integration error. In the evaluated implementation its unscaled charge variance is denoted R_q , so that $\text{var}(\nu_q) = \bar{\alpha}^2 R_q$. Treating the preceding ESR error as a zero-mean conditional uncertainty with variance P_{rr} yields

$$R_C \simeq R_{\Delta v} + (\Delta i_C)^2 P_{rr} + \hat{r}_C^2 R_{\Delta i} + \bar{\alpha}^2 R_q. \quad (16)$$

Here $R_{\Delta v} = 2R_v$, $R_{\Delta i} = 2R_i$, and R_q follows from current-noise variance, the ADC interval, and the integration length. Equation (16) propagates uncertainty; a biased $(r_C - \hat{r}_C) \Delta i_C$ remains bounded model mismatch rather than white noise.

D. Validity and Update Policy

An observation is accepted only in CCM with a valid timesamp, topology state, and complete safe window. Denoting

these prechecks by $\mathcal{S}_C$ and $\mathcal{S}_R$, the common evaluation stream implements

$$\begin{aligned} \mathcal{V}_C &= \text{CCM} \wedge \mathcal{S}_C \wedge (n_C > 0), \\ \mathcal{V}_R &= \text{CCM} \wedge \mathcal{S}_R \wedge (I_{\Sigma} \geq I_{\Sigma, \text{gate}}), \end{aligned} \quad (17)$$

with the predeclared $I_{\Sigma, \text{gate}} = 0.12$ A. Charge acceptance uses the CCM/safe-window flag and a nonzero accepted-charge count; the implementation has no separate numeric q_{gate} . A normalized-innovation-squared (NIS) threshold of 9 may reject additional TS-SLTVKE outliers. A failed gate holds the corresponding estimate. The evaluated-envelope minima in Section IV are proof constants, not online thresholds.

Since the capacitance regressor is $h_C = q/C_b$, low-charge rows inherently contribute little information. The positive information guarantee is therefore restricted to the evaluated valid-CCM envelope rather than asserted for arbitrarily small q .

IV. FINITE-WINDOW IDENTIFIABILITY AND INFORMATION BOUNDS

A. Information Structure

For accepted scalar rows, define $h_C = q/C_b$ and $h_R = k_R I_{\Sigma}$, with conditional variances R_C and R_R . Because the

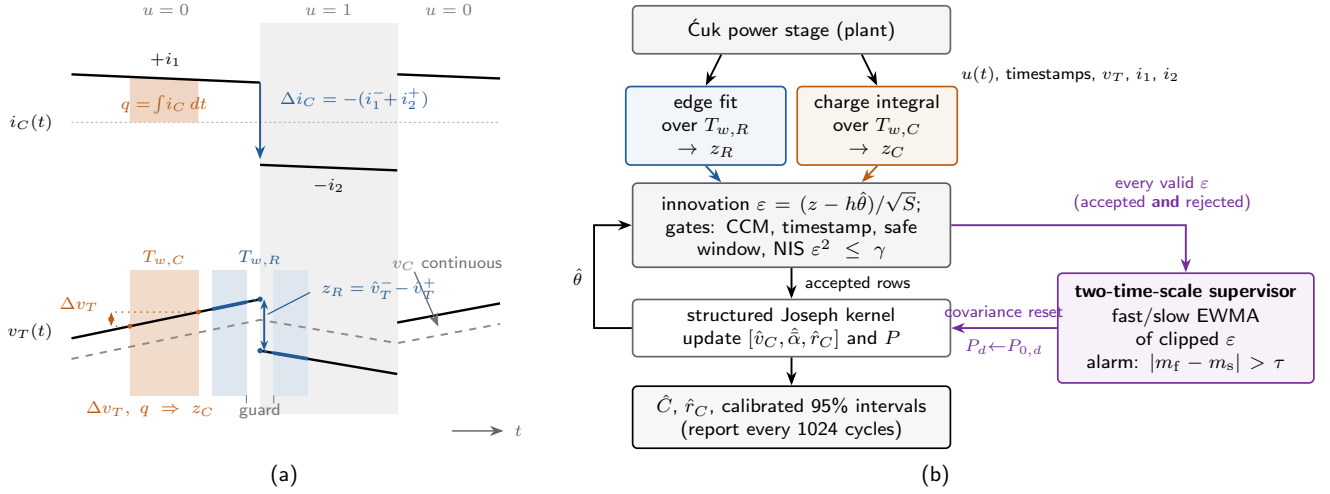


Fig. 2. Core method. (a) Topology-synchronous excitation and decoupled observations over 1.5 switching periods (schematic; guard exaggerated for readability): the capacitor current commutates between $+i_1$ and $-i_2$; the internal voltage v_C (dashed) is continuous while the terminal voltage v_T carries the ESR discontinuity, reconstructed by local fits in the guarded windows $T_{w,R}$ extrapolated to the common edge time; the charge integral q and endpoint difference Δv_T over $T_{w,C}$ form the capacitance row. (b) The complete supervised estimation loop: per-cycle features from the sensed converter waveforms form the two rows, the innovation ε is built from the fed-back estimate $\hat{\theta}$ and screened by the predeclared gates, accepted rows update the structured Joseph kernel, and the two-time-scale supervisor watches every valid ε —accepted or gate-rejected—restoring gain by a per-direction covariance reset after a sustained shift.

two observations act along disjoint parameter directions, the finite-window information Gramian is

$$\mathbf{G}_{\theta,N} = \sum_{k \in \mathcal{K}_C} \frac{1}{R_{C,k}} \begin{bmatrix} h_{C,k}^2 & 0 \\ 0 & 0 \end{bmatrix} + \sum_{k \in \mathcal{K}_R} \frac{1}{R_{R,k}} \begin{bmatrix} 0 & 0 \\ 0 & h_{R,k}^2 \end{bmatrix}. \quad (18)$$

Suppose every accepted charge interval has length at least $T_{w,C,\min}$ and satisfies $|i_C| \geq I_{C,\min}^{\text{eval}}$ without changing sign. Then $|q| \geq q_{\min}^{\text{eval}} = I_{C,\min}^{\text{eval}} T_{w,C,\min}$. If $R_C \leq R_{C,\max}$, $R_R \leq R_{R,\max}$, the calibrated k_R remains fixed with known conservative evaluated-envelope lower bound $k_{R,\min}^{\text{eval}} > 0$, and $I_\Sigma \geq I_{\Sigma,\min}^{\text{eval}} > 0$, windows containing m_C charge rows and m_R edge rows have the lower bounds

$$\underline{\mu}_C = m_C \frac{(I_{C,\min}^{\text{eval}} T_{w,C,\min} / C_b)^2}{R_{C,\max}}, \quad (19)$$

$$\underline{\mu}_R = m_R \frac{(k_{R,\min}^{\text{eval}})^2 (I_{\Sigma,\min}^{\text{eval}})^2}{R_{R,\max}}. \quad (20)$$

Proposition 1 (Operating-condition-induced finite-window identifiability): Assume valid CCM; k_R known and fixed during each identification window; sign-invariant accepted charge windows; positive accepted edge-current sum; finite conditional covariances; and at least one accepted charge and edge row in a finite window. Then

$$\mathbf{G}_{\theta,N} \succeq \text{diag}(\underline{\mu}_C, \underline{\mu}_R) \succ \mathbf{0}, \quad (21)$$

and (C, r_C) is locally identifiable over the accepted finite window.

Proof: Each charge row contributes a positive semidefinite matrix only in the $\bar{\alpha}$ direction, and each edge row contributes one only in the r_C direction. Applying the physical lower bounds in (19) and (20), summing m_C and m_R accepted contributions, and noting that both lower bounds are strictly

positive establishes (21). Since $C = C_b / \bar{\alpha}$ is locally one-to-one for $\bar{\alpha} > 0$, local identifiability transfers from $(\bar{\alpha}, r_C)$ to (C, r_C) . $\square$

The result is conditional: PWM alone does not guarantee persistent excitation (PE) at every load or conduction mode. The Cuk topology supplies the physical directions, while accepted rows certify when the bound applies. The superscript “eval” marks minima over the evaluated CCM envelope; these constants are neither online gates nor post-hoc tuning values.

B. Numerical Lower-Bound Check

Across 45 independent Model-B CCM cases, the weakest physical quantities were

$$\begin{aligned} I_{C,\min}^{\text{eval}} &= 0.330108 \text{ A}, & I_{\Sigma,\min}^{\text{eval}} &= 0.438504 \text{ A}, \\ q_{\min}^{\text{eval}} &= 0.660217 \text{ } \mu\text{C}. \end{aligned} \quad (22)$$

with $T_{w,C,\min} = 2.0 \text{ } \mu\text{s}$ and $k_{R,\min}^{\text{eval}} = 0.957654$. The resulting weakest information bounds were

$$\underline{\mu}_C = 0.217943, \quad \underline{\mu}_R = 880.008. \quad (23)$$

The minimum ratios of empirical information to the corresponding conservative bound were 1.02872 and 1.02334. Hence every retained case exceeded its computed bound, while the weakest case rather than a nominal case determines the reported constants. These existence bounds are complemented by the CRLB efficiency evaluation of Section VII, which quantifies how closely the realized estimator variance approaches the accepted-row information.

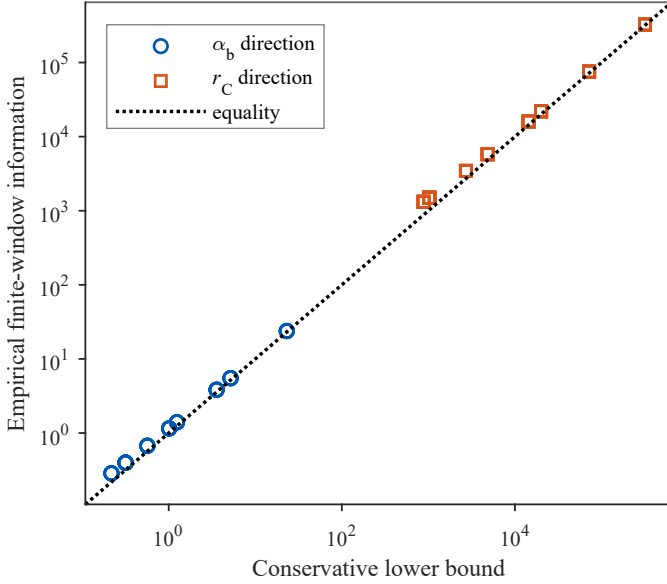


Fig. 3. Theoretical lower bounds versus empirical finite-window information over the Model-B CCM cases. Circles denote normalized inverse capacitance and squares denote physical ESR; their parameterization-dependent magnitudes and units are not directly comparable. All accepted cases lie above the equality line.

C. Covariance Contraction for the Uncertainty Extension

For the uncertainty extension, let P_n be a posterior variance, Q_N the process variance accumulated over a finite window, and $\mu \geq \underline{\mu} > 0$ the accepted information. Then

$$P_{n+1} \leq \frac{P_n + Q_N}{1 + \mu(P_n + Q_N)}. \quad (24)$$

Proposition 2 (Finite-window covariance boundedness): For recurring valid updates, finite $0 \leq Q_N \leq \bar{Q}_N$, and $\mu \geq \underline{\mu} > 0$, the posterior covariance satisfies (24). For constant $\bar{Q}_N$, its positive fixed point is

$$P^* = \frac{-Q_N + \sqrt{Q_N^2 + 4Q_N/\mu}}{2}. \quad (25)$$

For $Q_N = 0$, $P_n^{-1} \geq P_0^{-1} + n\mu$.

This conditional linear-model result is consistent with standard filtering theory [37], [38]; it is not a global nonlinear-converter convergence theorem. Appendix B gives the derivation and validated fixed-point values. Projection was inactive in 200 legal-PE seeds and is treated as a safety constraint rather than the source of contraction.

V. ONLINE ESTIMATION: KERNELS AND THE SUPERVISED-RESET REALIZATION

All realizations in this section consume the same accepted edge and charge rows from Section III, separating observation effects from recursive-kernel effects.

A. Kernel Baselines

The lightweight baseline TS-D-RLS uses one scalar RLS recursion per health direction. For direction $j \in \{C, R\}$, an accepted row $z_{j,k} = h_{j,k}\theta_{j,k} + \nu_{j,k}$ updates

$$K_{j,k} = \frac{P_{j,k-1}h_{j,k}}{\lambda + h_{j,k}^2 P_{j,k-1}}, \quad (26)$$

$$\hat{\theta}_{j,k} = \hat{\theta}_{j,k-1} + K_{j,k}(z_{j,k} - h_{j,k}\hat{\theta}_{j,k-1}), \quad (27)$$

$$P_{j,k} = \lambda^{-1}(P_{j,k-1} - K_{j,k}h_{j,k}P_{j,k-1}), \quad (28)$$

with forgetting factor $\lambda = 0.9975$; only the direction associated with an accepted observation is updated, validity gating rejects rows that fail the predeclared acceptance rules, and projection enforces the physical parameter intervals. The reported health values are $\hat{C} = C_b/\hat{\alpha}$ and $\widehat{\text{ESR}} = \hat{r}_C$.

The Kalman kernel TS-SLTVKE maintains the structured state $\mathbf{x}_k = [v_{C,k} \ \bar{\alpha}_k \ r_{C,k}]^T$ with the charge-driven transition

$$\mathbf{F}_k = \begin{bmatrix} 1 & q_k/C_b & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (29)$$

structured measurement directions $\mathbf{H}_V = [1, 0, 0]$, $\mathbf{H}_C = [0, q/C_b, 0]$, $\mathbf{H}_R = [0, 0, k_R I_\Sigma]$, scalar Joseph-form covariance updates, predeclared process and measurement covariances, and normalized-innovation-squared gating with the gate $\gamma = 9$. Its posterior covariance provides calibrated interval reporting. Both kernels are standard recursions [37], [38]; neither is claimed as a novelty, and the TS- names denote only their topology-synchronous, direction-specific use.

B. The Tracking–Confidence Tradeoff of the Unsupervised Kernels

The two kernels fail in complementary ways. The RLS baseline tracks and recovers quickly, but its auxiliary residual-weighted interval is uncalibrated (13.00% joint coverage in Section VII). The Kalman kernel reports calibrated intervals (88.93%), but after an abrupt health change every innovation exceeds the NIS gate, the update is rejected, the covariance stays small, and the estimate freezes at the stale value—the retained negative result of the unsupervised realization. The mechanism also identifies the cure: the gate-rejected innovations, which the point update discards, carry exactly the sustained same-sign evidence of a parameter change.

C. Primary Realization: Supervised-Reset Estimator TS-SRKE

TS-SRKE augments the Kalman kernel with a per-direction supervisor over the clipped normalized innovation of every valid row, accepted or rejected,

$$\varepsilon_{j,k} = \text{clip}\left(\frac{z_{j,k} - h_{j,k}\hat{\theta}_{j,k-1}}{\sqrt{S_{j,k}}}, \pm c\right), \quad S_{j,k} = h_{j,k}^2 P_j + R_{j,k}. \quad (30)$$

Two exponentially weighted means with separated time scales,

$$\begin{aligned} m_j^f &\leftarrow m_j^f + a_f(\varepsilon_{j,k} - m_j^f), \\ m_j^s &\leftarrow m_j^s + a_s(\varepsilon_{j,k} - m_j^s), \end{aligned} \quad (31)$$

act as the current innovation level and its bias reference. When $|m_j^f - m_j^s| > \tau$, the supervisor resets that direction's covariance to its initialization value, re-anchors $m_j^f \leftarrow m_j^s$, and holds off for N_h valid rows; the supervisor arms only after an N_w -row learning window per direction. All constants are predeclared and shared by both directions: $a_f = 1/16$, $a_s = 1/128$, $\tau = 2.5$, $c = 6$, $N_w = N_h = 32$.

The two-time-scale form is deliberate. Residual timing skew leaves a deterministic, near-periodic disturbance in the edge row whose σ -normalized amplitude grows as noise shrinks; a plain CUSUM misfires on it, whereas the fast filter attenuates the zero-mean disturbance below τ while a sustained parameter-change shift crosses it within a few valid rows. After a reset the Kalman gain is restored, the estimate re-converges as from initialization, and the reported interval widens and then re-tightens—an honest reflection of post-change uncertainty. The supervisor adds four multiplications and about seven additions/comparisons per valid row to the kernel; all components are classical [33], [36], and the construction is the contribution.

VI. VERIFICATION METHODOLOGY

A. Model and Data Hierarchy

The evaluation separates equation checks, independent plant traces, estimator comparisons, and device-realistic acquisition:

- 1) Model A is a switched-equation Ćuk model for sign, timing, and rapid dynamic checks.
- 2) Model B is an independently constructed Simscape model; 45 traces establish current scale, edge slope, and the blind observation data.
- 3) The estimator comparison uses 48 stratified blind operating/health cases, four noise profiles, and four residual timing-skew levels under common O1 observations.
- 4) The F28379D model adds register-realizable sampling, aperture/occupancy, reference/clock effects, analog-front-end (AFE) residuals, and Monte Carlo variation; it remains a simulation.

Supplementary closed-loop-regulation and light-load-availability studies reuse the locked pipeline on Model-A cycles, and a CRLB sensitivity study quantifies estimator efficiency; all three are reported in Section VII.

B. Factorial and Blind Comparisons

The observation–estimator factorial crosses a conventional mixed observation set (O0) and the proposed direction-decoupled set (O1) with RLS (E1), the dual extended Kalman filter (Dual EKF, E2), the structured LTV/Joseph estimator (E3), and the supervised-reset estimator (E4, TS-SRKE). All $2 \times 4 \times 768 = 6144$ paired rows are retained, and the E1–E3 rows reproduce the pre-supervision campaign exactly because the observation streams are seeded per condition and shared. The final comparison evaluates TS-SRKE against TS-D-RLS, TS-SLTVKE, and Dual EKF under O1 with predeclared primary-selection criteria: steady-state behavior within 2% of the Kalman parent, best-or-equal abrupt settling relative to the two retained realizations, coverage within two percentage

points of the parent, and no worse load-transient false-health peaks.

The broader baseline study includes closed-form, mixed-RLS, augmented/Dual-EKF, and Ćuk-adapted wavelet–KF methods. An NPC spectral method was excluded because the literature provides no unique mapping to the Ćuk edge/charge convention.

C. Fairness Rules and Metrics

All methods use identical blind cases, observations, seeds, initialization distributions, noise/timing realizations, and predeclared hyperparameters. Failed or saturated rows are retained, no method is retuned after blind evaluation, and paired effects use 10,000 bootstrap resamples.

Abrupt C/ESR discontinuities are estimator stress tests. The principal dynamic cases are linear and smooth C-down, ESR-up, and joint ramps lasting 0.1, 1, 10, and 100 s. Long ramps use trace-derived O1 sufficient statistics, while 0.1 s cases are cross-checked at every switching pair with Model A. Load, input-voltage, and duty transitions hold health constant to measure false-health response. None of these trajectories is hardware aging data.

Static endpoint errors are

$$e_C = 100 \left| \hat{C}_N / C - 1 \right|, \quad e_R = 100 \left| \hat{r}_{C,N} / r_C - 1 \right|, \quad (32)$$

with the mean absolute percentage error (MAPE) and 95th percentile (p95) computed over retained rows. “Convergence cycles” is the first cycle beginning 32 consecutive cycles with both relative errors within 5%; unconverged cases are assigned the 1024-cycle horizon. Rejected cycles hold the estimate and remain in the count. This internal settling metric differs from the 1024-cycle (20.48-ms) external reporting cadence.

For a trajectory $y(t)$ with initial truth y_0 , the implemented tracking metric is

$$\text{nRMSE}_y = \sqrt{\frac{\int (\hat{y} - y)^2 dt}{\int (y - y_0)^2 dt}}. \quad (33)$$

Combined nRMSE is the unweighted mean of finite C and ESR values. Normalized lag is the largest finite C/ESR delay from zero-mean cross-correlation and 50%-trajectory crossings, divided by trajectory duration. Missed threshold crossings are retained but are not used to select the primary estimator. Supplementary Note S3 records the complete threshold and bootstrap definitions.

VII. RESULTS

A. Attribution to the Observation Framework

The observation–estimator factorial provides the clearest attribution. Replacing O0 with O1 while retaining the E1 RLS kernel reduced mean capacitance MAPE from 7.0366% to 0.3727%, mean ESR MAPE from 1.1970% to 0.2233%, and mean internal convergence from 665.41 to 13.21 switching cycles. The paired ESR reduction was 0.97369 percentage points (95% bootstrap interval 0.92352–1.0265); all 4608 rows were retained. Supplementary Note S3 gives the other paired intervals.

TABLE V
BLIND VERIFICATION AND ESTIMATOR SETTINGS

Item	Predeclared setting	Role/source boundary
Health/load matrix	$C/C_b \in \{0.8, 0.9, 1.0\}$; $r_C/r_b \in \{1, 1.5, 2\}$; load factors 0.58/1/1.45	Stratified blind plant matrix; retained CCM rows only
Initialization	realized $\hat{C}_0/C \in [0.708, 1.287]$; $\hat{r}_{C,0}/r_C \in [0.512, 1.483]$	Read from the blind-case table
Parameter bounds	$C/C_b \in [0.65, 1.35]$; $r_C/r_b \in [0.35, 2.50]$	Predeclared projection limits; failed/saturated rows retained
Noise profiles	$(\sigma_v, \sigma_i) = (1 \text{ mV}, 0.5 \text{ mA}), (5 \text{ mV}, 2 \text{ mA}), (10 \text{ mV}, 5 \text{ mA}), \text{ and } (2.2 \text{ mV}, 1.2 \text{ mA})$	Shared realization; last pair is F28379D-realistic
Residual skew	0, 20, 50, and 100 ns	Shared by all compared estimators
Online validity/gating	valid CCM/state/time and safe feature; $I_{\Sigma, \text{gate}} = 0.12 \text{ A}$; no numeric q_{gate} ; TS-SLTVKE NIS gate 9	Fixed before evaluation, not inferred from PE minima
Evaluated PE constants	$I_{C, \min}^{\text{eval}} = 0.330108 \text{ A}$; $I_{\Sigma, \min}^{\text{eval}} = 0.438504 \text{ A}$; $q_{\min}^{\text{eval}} = 0.660217 \mu\text{C}$	Minima over 45 evaluated Model-B CCM cases; proof only, never online tuning thresholds
Blind comparison	48 cases $\times$ 4 noise levels $\times$ 4 skews; O0/O1 $\times$ RLS/Dual EKF/LTV-Joseph/TS-SRKE	O1 is then shared by all four estimators
Statistics/dynamics	10,000 paired bootstrap resamples; 0.1/1/10/100-s C, ESR, and joint ramps	Load, input-voltage, and duty steps hold physical health fixed
TS-SLTVKE settings	$Q_\alpha = 2 \times 10^{-9}$, $Q_r = 5 \times 10^{-9}$, $P_{0, \alpha} = 0.0144$, $P_{0, r}$ scale 0.45, NIS 9	Held fixed before blind comparison; shared by the TS-SRKE kernel
TS-SRKE supervisor	$a_f = 1/16$, $a_s = 1/128$, $\tau = 2.5$, clip 6, warm-up/hold-off 32 rows	Predeclared, shared by both directions; never retuned per case
Device acquisition	1.093 MS/s/module; 0.5- μs guard; $T_{w, R} = 2.2 \mu\text{s}$; three points/side	Constrained F28379D simulation only

TABLE VI
STATIC O1 ESTIMATOR COMPARISON WITH INTERNAL CONVERGENCE

Method	C MAPE (%)	ESR MAPE (%)	C/ESR p95 (%)	Conv. (int. cycles)
TS-D-RLS	0.3727	0.2233	1.3182/0.7002	13.21
TS-SLTVKE	0.3011	0.6446	1.1069/2.4992	33.09
Dual EKF	0.3043	1.1938	1.1492/3.8602	40.58
TS-SRKE	0.3011	0.6446	1.1069/2.4992	33.09

The TS-SRKE row equals its Kalman parent digit for digit: the supervisor stayed silent on all 768 static blind rows.

O1 also reduced capacitance MAPE by more than 6.7 percentage points for Dual EKF and E3 LTV/Joseph. The improvement was not universal: the Dual-EKF ESR-p95 effect was negative and its interval did not favor O1. Thus, the factorial supports the observation framework without implying monotonic improvement in every estimator metric.

B. Static Estimator Comparison Under O1

Table VI compares the four candidates on the $48 \times 4 \times 4$ O1 matrix; none diverged. The supervised TS-SRKE reproduced its Kalman parent *exactly* in every static metric—the supervisor never fired across the 768 blind rows, including the low-noise/high-skew cells that carry the largest deterministic edge disturbance. TS-D-RLS retains the lowest ESR error and fastest native convergence, a baseline strength reported throughout. Under the predeclared selection criteria of Section VI, TS-SRKE is adopted as the primary realization once the dynamic results below are included; TS-D-RLS is retained as the lightweight tracking alternative.

C. Noise and Timing Robustness

All combinations of four noise profiles and 0, 20, 50, and 100 ns residual mismatch were retained. At 50 ns

under F28379D-realistic noise, C/ESR p95 errors were 0.4964%/0.3385% for TS-D-RLS, 0.4132%/0.6888% for TS-SLTVKE, and 0.4211%/1.9141% for Dual EKF; the TS-SRKE rows coincide with TS-SLTVKE at every noise/skew cell. Figure 6 visualizes that device-realistic profile; C and ESR percentiles are shown separately because their arithmetic mean has no independent interpretation.

D. Efficiency Relative to the Conditional CRLB

A conditional CRLB was computed from the accepted-row information of the structured estimator lineage (Section V) on Model-A traces through the register-nominal measurement chain, over fifteen load, duty, ADC-rate, AFE-bandwidth, and timing sensitivity cases with twenty seeds each. When the chain matches its commissioned configuration, the empirical RMSE stayed within factors 1.38–2.34 (capacitance) and 1.35–3.27 (ESR) of the bound; the medians over all fifteen cases were 2.17 and 2.40 (Fig. 7). Deviations from the commissioned chain instead appear as bias that the noise-only bound does not model: operating at a reduced 1.5-MHz AFE bandwidth raised the capacitance ratio to 25.0, an increased 3-MHz bandwidth raised the ESR ratio to 10.2, and 100/200-ns residual timing mismatch raised the capacitance ratio to 13.0/39.6. The bound therefore certifies near-efficient noise behavior at the commissioned configuration and flags calibration or timing bias away from it.

E. Abrupt Recovery: Resolving the Retained Failure

Abrupt discontinuities are recovery stress tests, not aging events. The unsupervised TS-SLTVKE did not satisfy the convergence criterion within 257 post-step cycles for either health step (tail errors 19.2911% after $C \rightarrow 0.8C$ and 49.9384% after $r_C \rightarrow 2r_C$): once the parameter jumps, every innovation

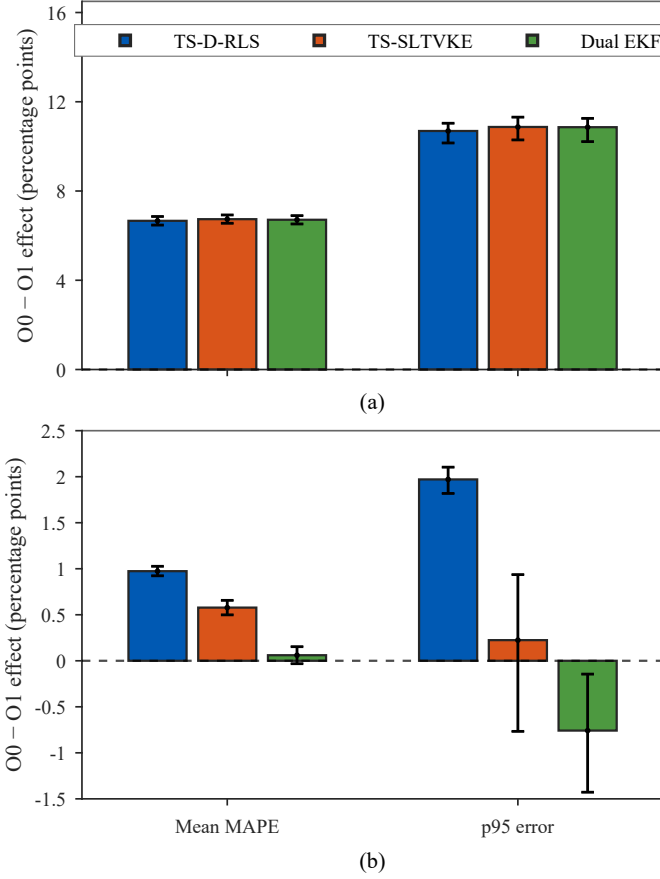


Fig. 4. Paired O0-O1 effects on MAPE and p95 error with bootstrap intervals: (a) capacitance; (b) ESR. Positive values favor the topology-decoupled O1 observations; the adverse Dual-EKF ESR-p95 effect is retained.

TABLE VII
ABRUPT-STEP SETTLING ON FROZEN STRESS STREAMS

Method	$C \rightarrow 0.8C$ (ms)	$r_C \rightarrow 2r_C$ (ms)	joint (ms)
TS-D-RLS	10.72	17.68	17.28
TS-SLTVKE	64.72	no recovery	no recovery
Dual EKF	61.00	0.24	61.36
TS-SRKE	1.52	0.48	1.20

exceeds the NIS gate and the estimate freezes. Table VII and Fig. 8 show the supervised resolution on the frozen Model-A stress streams. TS-SRKE settles the capacitance, ESR, and joint steps in 1.52, 0.48, and 1.20 ms—one to two external report frames—versus 10.72–17.68 ms for TS-D-RLS and no recovery for its own parent. Dual EKF settles the ESR step fastest (0.24 ms) because it carries no gate, but its uncalibrated coverage, worst false-ESR transients, and highest cost are unchanged, so it remains excluded by the companion criteria.

F. Slow Degradation Tracking

Across the 0.1–100 s sweep in Table VIII, TS-D-RLS had lower aggregate combined nRMSE and normalized lag at all four durations, and the supervised TS-SRKE tracked its Kalman parent to within rounding: slow drift stays below the supervisor threshold, so the resets that cure abrupt

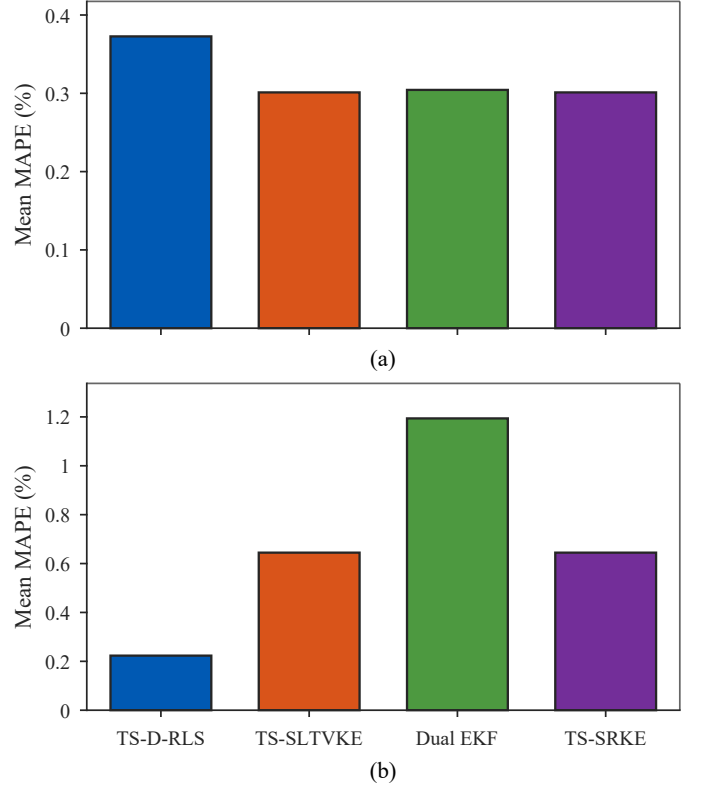


Fig. 5. Static mean MAPE under common O1 observations: (a) capacitance; (b) ESR. The TS-SRKE bars coincide with the TS-SLTVKE bars because the supervisor is silent in steady state.

TABLE VIII
AGGREGATE SLOW-DEGRADATION TRACKING
(M1 = TS-D-RLS, M2 = TS-SLTVKE, M4 = TS-SRKE)

Ramp (s)	Combined nRMSE			Normalized lag (%)		
	M1	M2	M4	M1	M2	M4
0.1	0.1161	0.2160	0.2160	8.053	22.027	22.027
1	0.0163	0.0454	0.0454	1.360	3.537	3.537
10	0.0160	0.0456	0.0449	1.752	4.492	4.492
100	0.0213	0.0477	0.0478	2.827	4.868	4.868

steps deliberately do not engage on health-realistic ramps (isolated firings marginally improved the 10-s aggregate). The lightweight baseline therefore keeps a genuine ramp-tracking advantage, reported as such. Individual trajectories can still favor the Kalman side; one 10-s C-ramp case did so.

G. Operating-Point Transient Immunity

Load, input-voltage, and duty transitions held health fixed. TS-SLTVKE reduced the mean false-C peak (0.7570% versus 1.3017%), whereas TS-D-RLS reduced the mean false-ESR peak (1.8474% versus 5.1919%) and recovered faster. The supervised TS-SRKE reproduced its parent on every transient row—operating transients did not trigger a single reset, so the supervisor adds no false-health response of its own. Neither kernel family dominates operating-transient immunity; Supplementary Fig. S4 shows the retained load-step traces.

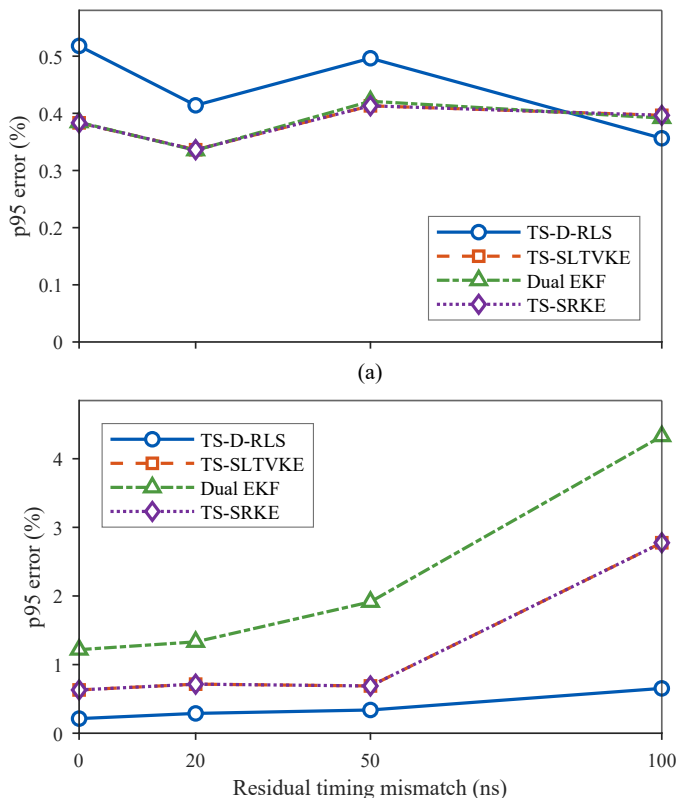


Fig. 6. p95 errors versus residual timing mismatch under the F28379D-realistic noise profile and shared O1 observations: (a) capacitance; (b) ESR. The TS-SRKE curve overlays its Kalman parent exactly; distinct line and marker styles preserve method identity in grayscale.

H. Closed-Loop Operation

All preceding rows hold the duty command open loop. A supplementary study closed a deliberately simple, low-bandwidth discrete PI voltage loop with 0.5% duty quantization around the Model-A plant and re-commissioned the edge gain under feedback ($k_R^{cl} = 0.9987$); this and the light-load study below evaluate the observation pipeline with the lightweight TS-D-RLS realization. Three blind health cases then applied a $1.45\times$ load step, a $+20\%$ input-voltage step, and a 16-to-12-V reference step at 30 ms. Steady-tail capacitance and ESR errors remained at or below 0.222% and 0.111%, internal convergence took 3–7 cycles, and the largest post-transition excursions were 0.410% in capacitance and 0.102% in ESR (Fig. 10). During the input-voltage transient, 15.7% of cycles failed the CCM gate and were rejected with the estimate held; accuracy was unaffected. The study uses one noise seed per case and does not evaluate controller design itself.

I. Availability Toward the Light-Load Boundary

Because identifiability is conditional on valid CCM, a fixed-duty sweep quantified observation availability from 100% down to 6.7% load with the same locked pipeline. Steady-state cycle acceptance remained 100% down to 20% load, fell to 68.8%, 49.8%, 36.1%, and 9.9% at 16.7%, 14.3%, 12.5%, and 10.0% load, and reached zero at and below 8.3%

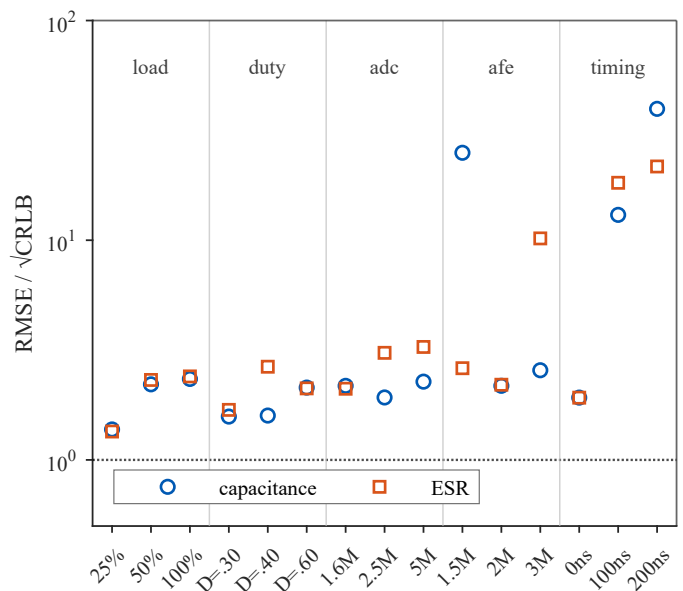


Fig. 7. Empirical RMSE relative to the conditional CRLB over fifteen sensitivity cases (twenty seeds each), grouped by dimension. The bound is conditional on accepted rows and models noise only, so ratios far above one indicate calibration or timing bias rather than estimator inefficiency.

TABLE IX
PER-ACCEPTED-OBSERVATION COMPLEXITY

Method	Mult.	Add.	Div.	Est. time (μ s)
TS-D-RLS	28	20	1	1.36
TS-SLTVKE	46	34	1	2.12
Dual EKF	62	47	2	3.05
TS-SRKE	50	41	1	2.30

load, where the estimate is held (Fig. 11). Internal convergence lengthened from 5 to 39 cycles over the fully available range, and endpoint errors over accepted rows stayed at or below 1.4% down to 10% load. The switched Model-A equations flag would-be DCM cycles but do not emulate DCM conduction [39]; the sweep therefore quantifies availability, not DCM-mode accuracy.

J. Complexity and Uncertainty

On the common arithmetic basis in Table IX, TS-D-RLS, TS-SLTVKE, Dual EKF, and TS-SRKE require 28, 46, 62, and 50 multiplications per accepted observation; the supervisor costs four multiplications and about seven additions/comparisons on top of its kernel. The joint 95% interval coverages are 13.00%, 88.93%, 81.90%, and 88.93%, respectively. TS-SRKE therefore carries its parent's calibrated confidence for four extra multiplications per row, and TS-D-RLS remains the lowest-cost tracking realization. The listed times are arithmetic estimates, not measured target worst-case execution time (WCET); acquisition and feature-processing costs are separate.

VIII. F28379D DEVICE-REALISTIC FEASIBILITY

The device-realistic study tests whether the internal ADC architecture can acquire the proposed observations under

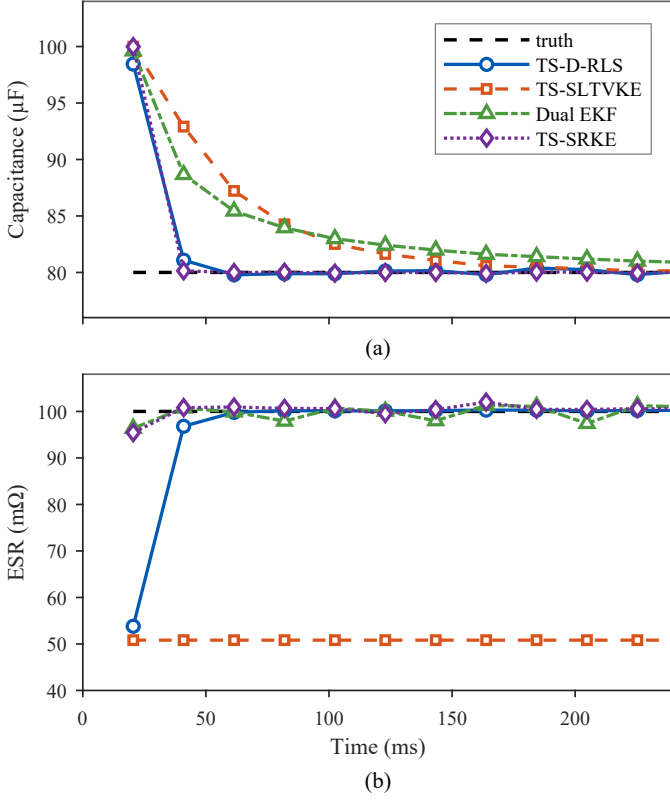


Fig. 8. Report-cadence recovery after the abrupt health steps on identical frozen streams: (a) capacitance estimates after $C \rightarrow 0.8C$; (b) ESR estimates after $r_C \rightarrow 2r_C$. The unsupervised TS-SLTVKE freezes at the stale value; TS-SRKE re-enters the band within one to two report frames.

TABLE X
F28379D DEVICE-REALISTIC SUMMARY

Item	Specified value
ADC architecture	four modules, 16-bit differential
Aperture/rate	320 ns / 1.093 MS/s per module
Edge acquisition	0.5-μs guard; $T_{w,R} = 2.2$ μs; 3 points/side
AFE boundary	differential path; edge-band CMRR ≥ 70 dB
Health reporting	1024 cycles (20.48 ms)
Worst p95 C/ESR	1.6425% / 2.6109%

register-realizable timing and an explicit high-common-mode analog front end (AFE). Four synchronized 16-bit differential ADC modules use a 320-ns aperture and sustain 1.093 MS/s per module. A one-edge schedule places three samples on each side of the reconstructed edge within $T_{w,R} = 2.2$ μs after a 0.5-μs guard. Direct ADC connection is infeasible over the simulated common-mode range, so feasibility requires the specified differential AFE, edge-band common-mode rejection, external reference, clock, and calibration residual.

Seven representative conditions with 200 Monte Carlo seeds each yielded worst-case p95 capacitance and ESR errors of 1.6425% and 2.6109%; every condition met its predeclared threshold. Table X summarizes the evidence needed for the feasibility claim. Register timing, aperture margin, DMA traffic, utilization, AFE component assumptions, common-mode extrema, and coverage ranges are retained in Supplementary Note S1.

The 1.36-μs TS-D-RLS value is an estimator-arithmetic estimate, not platform WCET. Firmware compilation, linker placement, measured execution time, and bench AFE verification remain open; the result supports a constrained simulation architecture rather than hardware validation.

IX. DISCUSSION

A. Why Observation Design Matters

The O0–O1 factorial separates observation design from estimator complexity: with the RLS kernel unchanged, O1 reduced both MAPE values and the convergence time by the factors reported in Section VII. This attribution agrees with the circuit physics: the switching edge exposes the ESR drop, while the in-state charge balance exposes inverse capacitance. More elaborate recursion cannot reliably compensate for observations that mix these effects or compare mismatched timestamps.

The scoped literature matrix contains no disclosed method combining the Ćuk transfer-capacitor target, $+i_{L1} \leftrightarrow -i_{L2}$ commutation, timestamp-reconstructed edge information, and direction-specific charge/edge updates. This is a bounded review finding, not a universal first-in-literature claim.

B. Why TS-SRKE Is Primary and What TS-D-RLS Retains

TS-SRKE satisfies all four predeclared primary criteria: it reproduces its Kalman parent digit for digit on every static, noise/timing, and operating-transient row (zero supervisor firings outside genuine health changes), keeps the parent’s calibrated 88.93% joint coverage, and settles abrupt health steps in one to two report frames where the parent freezes and the RLS baseline needs an order of magnitude longer. The supervisor thus removes the single disqualifying failure of the calibrated realization at four extra multiplications per row. The choice does not imply universal superiority: TS-D-RLS retains the lowest static ESR error, the fastest native convergence, the better aggregate ramp tracking at every duration, and the lowest cost, and it remains the recommended realization when only point tracking under smooth degradation is required. Dual EKF settles the ESR step faster because it carries no gate, but its uncalibrated coverage and worst false-ESR transients keep it excluded.

C. Limitations

The evidence has seven principal boundaries.

- 1) Identifiability and updates require valid CCM; DCM and weak-excitation intervals hold the estimate, with availability quantified toward the light-load boundary in Section VII. The method assumes availability of both inductor currents; applications lacking either measurement require an additional sensor or independently validated current reconstruction.
- 2) The reported ESR is an effective condition indicator at the modeled switching, thermal, and frequency conditions, not a temperature-independent aging variable.
- 3) The 0.1–100 s trajectories are synthetic or trace-derived bandwidth tests, not run-to-failure data; abrupt discontinuities are recovery stress tests rather than aging events.

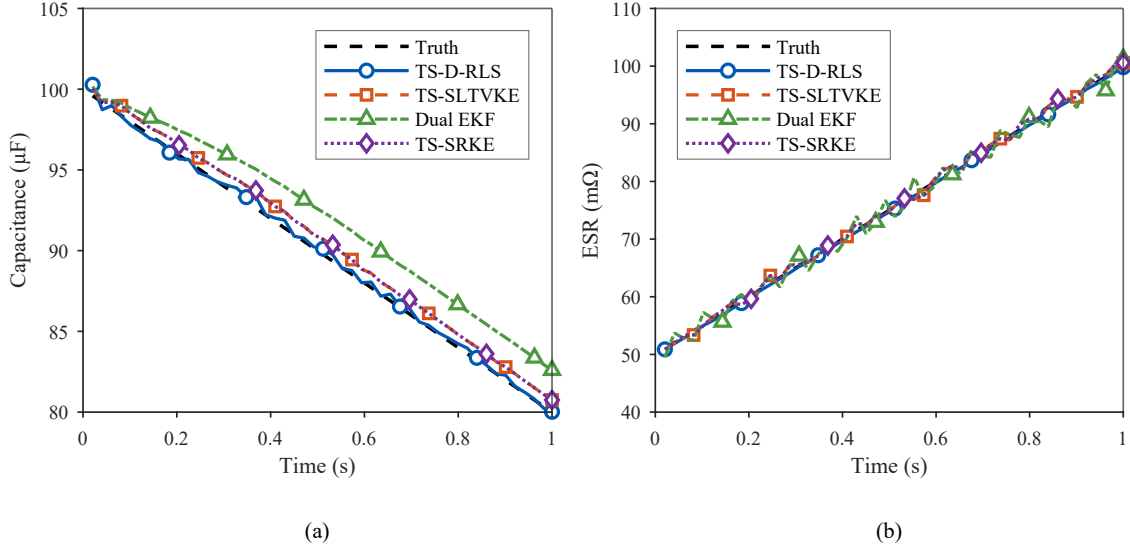


Fig. 9. Representative joint health ramps: (a) capacitance-down and (b) ESR-up trajectories with the planted truth. Long-duration trajectories use trace-derived O1 sufficient statistics rather than full Simscape transients or measured aging data.

- 4) Hardware validation remains open for edge ringing, isolation/probing, AFE common-mode rejection, and production calibration.
- 5) The F28379D study is device-realistic simulation; target compilation and measured WCET have not been demonstrated.
- 6) The supervisor resolves the abrupt-recovery side of the former tracking–confidence tradeoff, but TS-D-RLS still tracks smooth ramps better and the auxiliary TS-D-RLS intervals remain under-covered; the supervisor’s components are classical, and its constants, while predeclared and shared across all cases, were designed against the evaluated disturbance classes.
- 7) The closed-loop and light-load supplements use the Model-A plant with one noise seed per case, one deliberately simple PI regulator, and the TS-D-RLS realization; they do not evaluate controller-interaction breadth, DCM-mode operation, or the supervised estimator under feedback.

D. Experimental Roadmap

The next stage should use a controlled C/ESR network in the energy-transfer branch, the specified F28379D schedule, separate absolute- and edge-voltage paths, and synchronized inductor-current measurements. Tests should sweep input voltage, duty ratio, load, and temperature; characterize ringing and AFE rejection; verify calibration against offline impedance measurements; and measure WCET before firmware release.

X. CONCLUSION

This paper converted the normal bidirectional energy transfer of a Ćuk converter into two topology-synchronous capacitor-health observations—a timestamp-reconstructed switching-edge row for ESR and an ESR-compensated safe-window charge row for capacitance—and

proved positive finite-window information bounds that establish local C–ESR identifiability under valid CCM gates. On these observations, the supervised-reset estimator TS-SRKE closed the tracking–confidence gap of the unsupervised kernels: it reproduced its Kalman parent exactly in steady state with 88.93% joint interval coverage, and settled abrupt capacitance and ESR steps in 1.52 and 0.48 ms where the parent froze, at four extra multiplications per row.

On the blind matrix, replacing the conventional mixed observation with the proposed set reduced capacitance MAPE from 7.0366% to 0.3727% with the RLS kernel unchanged—the observation design, not estimator complexity, carries the largest gain. TS-SRKE then achieved 0.3011%/0.6446% C/ESR MAPE, while the lightweight TS-D-RLS baseline retained the best static ESR error, ramp tracking, and cost, and remains recommended where only smooth-degradation point tracking is required. Supplementary studies preserved pipeline accuracy under closed-loop regulation, quantified availability down to the light-load boundary, and showed near-CRLB noise efficiency; a constrained F28379D simulation supported device-realistic feasibility with worst p95 errors of 1.6425%/2.6109%.

Hardware and accelerated-aging studies must still validate AFE common-mode behavior, ringing, temperature dependence, calibration transfer, and measured execution time before experimental or lifetime claims are warranted.

APPENDIX A SWITCHED EQUATIONS UNDER THE ADOPTED SIGN CONVENTION

For completeness, the first-order switched equations used to verify the observation signs retain the transfer-capacitor ESR and set the other parasitics to zero. Here v_C is the ideal capacitor voltage and $v_T = v_C + r_C i_C$. With $u = 1$ (main

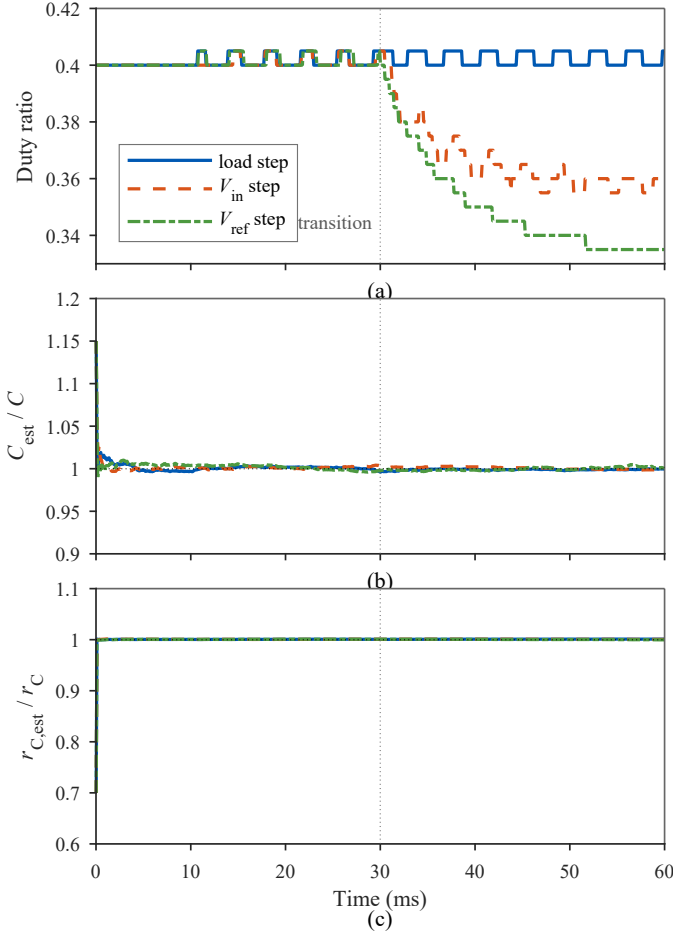


Fig. 10. Closed-loop supplement: (a) regulator duty responses to the load, input-voltage, and reference transitions at 30 ms; (b), (c) normalized capacitance and ESR estimates. Cycles failing the CCM gate during the input-voltage transition are rejected and hold the estimate.

switch on),

$$\dot{i}_1 = V_{in}/L_1, \quad \dot{i}_2 = (v_C - r_C i_2 - v_o)/L_2, \quad (34)$$

$$\dot{v}_C = -i_2/C, \quad \dot{v}_o = i_2/C_o - v_o/(RC_o). \quad (35)$$

With $u = 0$ (main switch off),

$$\dot{i}_1 = (V_{in} - v_C - r_C i_1)/L_1, \quad \dot{i}_2 = -v_o/L_2, \quad (36)$$

$$\dot{v}_C = i_1/C, \quad \dot{v}_o = i_2/C_o - v_o/(RC_o). \quad (37)$$

These equations yield (2) and match the specified Model-A implementation after its optional inductor, switch, diode, and source parasitics are set to zero. The ESR signs differ between states because $i_C = +i_1$ for $u = 0$ and $i_C = -i_2$ for $u = 1$.

APPENDIX B SCALAR COVARIANCE ENVELOPE

Let the posterior scalar variance at the start of a finite window be P_n . Process accumulation gives $P_n^- \leq P_n + Q_N$. Assimilating total scalar information μ gives

$$P_{n+1} \leq \left(\frac{1}{P_n + Q_N} + \mu \right)^{-1}, \quad (38)$$

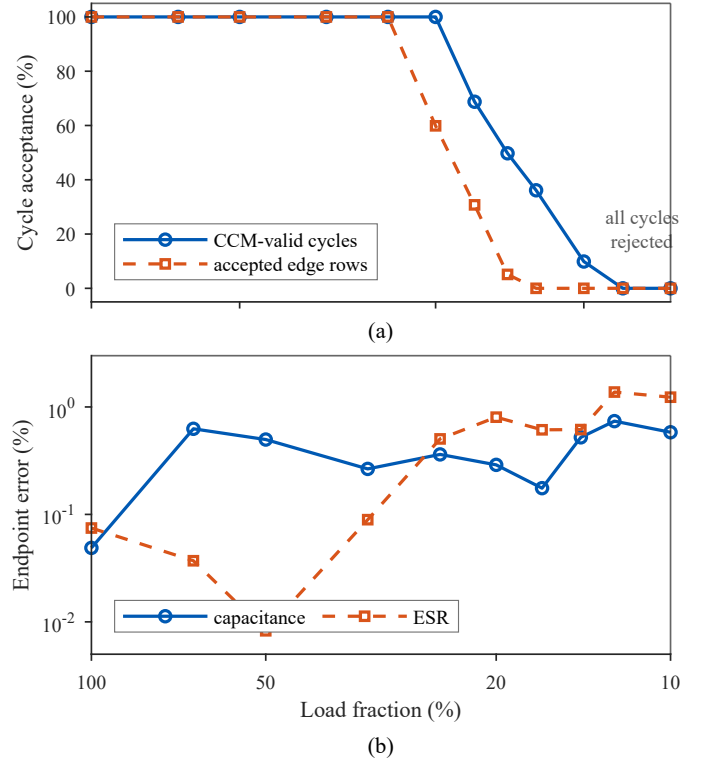


Fig. 11. Light-load availability sweep at fixed duty: (a) steady-state CCM-valid cycle and accepted-edge-row fractions; (b) endpoint errors over accepted cycles. Below the availability knee the gates reject all cycles and the estimate is held.

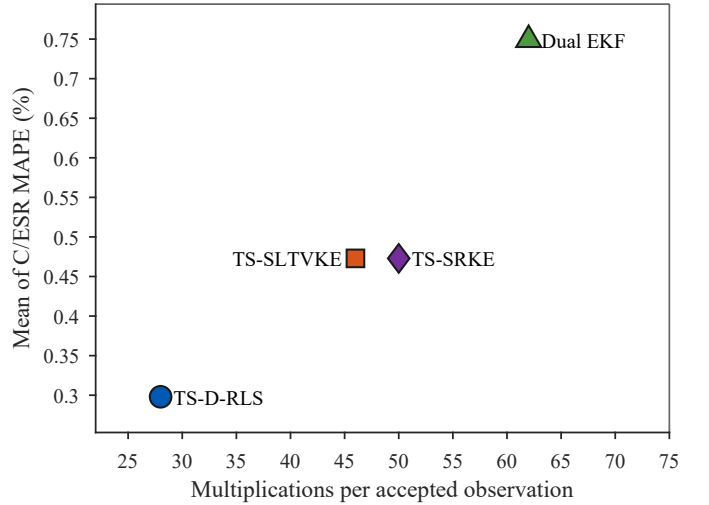


Fig. 12. Accuracy-complexity tradeoff under the common operation-count model. The ordinate is the arithmetic mean of C and ESR MAPE values, used only as a visual Pareto summary and not as the estimator-selection objective.

which is (24). Setting $P_{n+1} = P_n = P^*$ gives

$$\mu(P^*)^2 + \mu Q_N P^* - Q_N = 0, \quad (39)$$

whose positive root is (25). The map is increasing and bounded above by $1/\mu$. When $Q_N = 0$, inversion gives $P_{n+1}^{-1} \geq P_n^{-1} + \mu$ and hence $P_n \leq (P_0^{-1} + n\mu)^{-1}$. With bounded conditional mismatch, a finite case-dependent error term must be added.

The largest validated fixed points were 0.0112191 in the

normalized-capacitance direction and 8.82437×10^{-6} in the ESR direction.

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